# Colegio Connect

## A Unified Opportunity Platform for the UPRM Community

**Capstone Project Proposal**

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# 1. Introduction

## 1(a) Background and Motivation

Students at the University of Puerto Rico at Mayagüez (UPRM) have access to many academic and professional opportunities: jobs, internships, COOP positions, research positions, research-participation studies, and capstone sponsorship. These opportunities are published by different offices, departments, faculty members, employers, and external platforms. As a result, a student must already know where to look before finding them.This matters because these opportunities have real value for students. Internship experience has been shown to increase the interview-request rate of recent graduates by about 14% [1], and undergraduate research experiences increase students' understanding of research and their interest in graduate study [2]. A student who does not find these opportunities in time cannot benefit from them. Colegio Connect is a web platform that brings these opportunities together in one place for the UPRM community. Verified students discover and respond to opportunities, verified UPRM faculty and offices publish institutional opportunities, approved external companies publish authorized opportunities, and administrators moderate the platform. It also includes a two-sided capstone sponsorship path, where companies can publish project calls and student teams can publish projects that need support.

## 1(b) Justification

The need for this project is supported by published evidence and by a preliminary stakeholder assessment.

**Evidence from the literature.**

- **Awareness is a barrier.** Fouad et al. found that only about half of the students in their study knew about the career services offered by their institution, and fewer had used them [3]. Having a service available does not guarantee that students find it.
- **Opportunities affect outcomes.** Internships improve employment prospects [1], and research experiences support students' academic and professional development [2]. Missing these opportunities has a real cost.
- **Trust is a growing concern.** The Federal Trade Commission (FTC) has warned that job scams target college students by impersonating people and institutions they recognize [4]. Reported losses to job and employment-agency scams rose from $90 million in 2020 to $501 million in 2024 [5]. These are national figures, but they show that verification and moderation are necessary features, not optional ones.

**Preliminary stakeholder assessment.** Table 1 summarizes the needs of each stakeholder group based on the current UPRM environment. It will be refined with advisor and stakeholder feedback during the project.

| Stakeholder | Need | How Colegio Connect responds |
|---|---|---|
| Students | Find different types of opportunities in one place, know who posted them, and control their personal information. | One feed with search, filters, saving, and responses. Companies cannot browse student profiles. |
| UPRM Faculty / Offices / Departments | Publish opportunities in a structured way that students recognize as official. | Verified institutional role, structured listings, and visible source labels. |
| External Companies | Reach verified UPRM students without unrestricted access to student data. | Approved companies publish listings and see only the students who respond to them. |
| Capstone Teams and Sponsors | Find each other for real projects while protecting student contact information. | Company project calls and student projects seeking support; contact is shared only after the team accepts. |
| Admins / Moderators | Approve organizations, review reports, and keep decisions accountable. | Approval queues, moderation tools, account restrictions, and audit records. |

**Table 1. Preliminary stakeholder assessment.**

# 2. Problem Statement

## 2(a) Problem Definition and Capstone Relevance

**The problem this project solves is that UPRM opportunities are spread across separate channels, with no single trusted place to find them.** The Placement Office promotes jobs, internships, and COOP positions [6]; ADEM directs its students to Handshake [7]; research openings appear on individual department and laboratory pages [8], [9]; and industry sponsors propose engineering projects through Open Capstone [10]. Each channel works for its own purpose, but a student who wants a complete view must monitor all of them. Publishers face the same problem from the other side: they must guess which channels will reach the students they want. Opportunities that come from outside UPRM add a trust problem. Official UPRM pages are easy to recognize, but offers shared through email, social media, or unknown organizations are not, and scams targeting students are increasing [4], [5].

> **Problem statement.** UPRM opportunities are distributed across separate channels, and there is no single platform that combines their discovery, structured publication, verification, and moderation for the UPRM community.

**Capstone relevance.** Solving this problem requires much more than a page of links. Colegio Connect is a multi-user software system that needs authentication, role-based authorization, a relational data model, search and filtering, organization approval, moderation workflows, notifications, privacy controls, and measurable performance and reliability targets. These components must be designed, built, tested, and documented together, which makes the project an appropriate scope for a Computer Science and Engineering Capstone.

## 2(b) Target Domain, Client, Users, and Limitations

**The target domain of this solution is** the discovery of academic and professional opportunities within the UPRM community.

**The target client is the UPRM community,** with UPRM students as the primary users. The other users are the people and organizations that publish opportunities for them and the administrators who keep the platform safe. Table 2 describes the four user roles.

| Role | Who | Main capabilities |
|---|---|---|
| Student | Verified UPRM students (institutional email). | Browse, search, filter, save, and respond to listings; report listings; publish capstone projects seeking support; accept or decline sponsor interest. |
| UPRM Faculty / Office / Department | Professors, researchers, departments, and offices with an authorized institutional role. | Publish research, research-participation, departmental, and capstone opportunities, labeled as UPRM sources; view respondents to their own listings. |
| External Company | Employers, recruiters, and sponsors approved by administrators. | Publish jobs, internships, COOP positions, and capstone calls; view only respondents to their own listings; send Express Interest to student projects. |
| Admin / Moderator | Authorized UPRM members. | Approve companies and privileged roles, review reports, hide or remove listings, restrict accounts, and review audit history. |

**Table 2. User roles.**

**Privacy rule.** Companies cannot search or browse students. A company sees a student's information only when that student responds to one of its listings, or when a student team accepts its Express Interest request on a capstone project.

**Limitations of the solution.**

- **Coverage depends on participation.** The platform cannot guarantee that every UPRM opportunity will appear on it.
- **Verification reduces risk but is not a guarantee.** Approval, reporting, and moderation lower the chance of fraudulent listings but cannot confirm that every listing is accurate.
- **It does not replace existing processes.** Employer application systems, CPSHI review of human-subject research [11], Open Capstone agreements, and other university processes remain under their current owners. Colegio Connect links to them.
- **It is an independent student project.** It is not an official UPRM service unless UPRM endorses it, and it must work even if no specific office participates.
- **It is a prototype.** It is evaluated against the test workload in Section 3(b), not as a production campus service.
- **Out of scope:** AI recommendations, interview scheduling, in-app messaging, automated scam detection, appeal workflows, and direct integration with UPRM information systems.

## 2(c) Proposed Innovation

**The innovation of this proposal is the integration of separate opportunity types into one platform with a single trust and privacy model.** Job boards, research portals, and capstone programs already exist separately at UPRM [6], [7], [10], and previous CSE Capstone projects, PRJobs [12] and the UPRM Research Portal [13], addressed jobs and research individually. Colegio Connect does not claim that any one of these features is new. What is new is combining them:

1. **One discovery experience for six opportunity types.** Jobs, internships, COOP positions, research positions, research-participation studies, and capstone opportunities share search, filters, saving, reporting, and notifications, while each type keeps its own required fields.
2. **One trust and privacy model.** Students know that a company passed an approval process, companies know that respondents are verified UPRM members, and companies never receive open access to the student population.
3. **Two-sided capstone sponsorship.** Companies can publish project calls, and student teams can publish existing projects and describe the support they need, such as mentorship, equipment, funding, data, or facilities. Companies can express interest, but contact is shared only after the team accepts. Online sponsor matching has been used for capstone courses before [14], but not combined with other opportunity types in one campus platform.

The system is built on a shared foundation for accounts, roles, and listings, so new opportunity types can be added later without redesigning it.

# 3. Project Objectives

Colegio Connect will be developed as a working web prototype during an eight-week implementation period, completed before the end of the Fall 2026 semester. All objectives are software objectives: each one describes a function or quality of the system that can be tested, has a numeric target, and has a deadline within the eight weeks.

## 3(a) Project Objectives

| ID | Objective | Deadline | Indicators |
|---|---|---|---|
| O1 | **Unified discovery.** Implement one feed where signed-in users can browse, search, filter, and open listings from all six opportunity categories, and verified students can save them, with discovery requests answered within 2 seconds under the test workload. | Week 4 | I1–I4 |
| O2 | **Access control and privacy.** Implement email verification and server-side authorization so that every protected action checks the user's role, ownership, approval status, account status, and consent, with zero unauthorized disclosures of student data. | Week 6 | I5–I7 |
| O3 | **Moderation and accountability.** Implement reporting, listing review, hiding and removal of listings, account restrictions, and an audit log that records every administrative action, working even when no UPRM office participates. | Week 6 | I8–I9 |
| O4 | **Two-sided capstone sponsorship.** Implement the company-originated route (company call, student response) and the student-originated route (team project, Express Interest, accept or decline), with no private contact information released before acceptance. | Week 8 | I10–I12 |
| O5 | **Reliability, security, and accessibility.** Deliver a deployed build that passes the final acceptance suite, security scan, accessibility checks, recovery test, and usability test with the targets in Section 3(b). | Week 8 | I13–I17 |

**Table 3. Project objectives.**

## 3(b) Measurable Performance Indicators

Each indicator is specific, measurable, achievable within the project, relevant to one objective, and time-bound (SMART). An indicator is first tested in the sprint that builds the feature and again at final acceptance in Week 8; the full target must be met in Week 8.

**Test workload.** Performance tests use a documented synthetic dataset of at least 5,000 listings, 1,000 user accounts, and 50 organizations, with 25 concurrent virtual users. This is a repeatable test workload, not an estimate of real UPRM traffic. The hosting environment will be selected by the end of Week 2, and the final report will record its CPU, memory, test duration, response times, and error rate.

**Table 4. Measurable performance indicators.**

| ID | Indicator and target | How measured | When |
|---|---|---|---|
| I. 1 | All 6 categories can be created, published, found, filtered, and opened. | Automated end-to-end tests | Weeks 4, 6, 8 |
| I. 2 | At least 18 predefined search and filter cases return only matching records (100% correct). | Automated tests | Weeks 4, 6, 8 |
| I. 3 | 95% of discovery requests complete within 2 seconds, with a server error rate below 1%, under the test workload. | Load test (k6) | Weeks 4, 6, 8 |
| I. 4 | Save, unsave, and revisit work in all 6 categories. | Automated tests | Weeks 4, 6, 8 |
| I. 5 | 100% of registration attempts with non-UPRM emails and invalid or expired codes are rejected, and verification requests are rate limited. | Automated tests | Weeks 2, 4, 6, 8 |
| I. 6 | 100% of at least 10 negative-authorization tests pass (e.g., unapproved company publishing, cross-organization access, suspended-account actions, non-team member accepting sponsorship). | Automated integration tests | Weeks 2, 4, 6, 8 |
| I. 7 | 0 private student fields appear in API responses during browsing, searching, saving, or unauthorized requests. | Automated API-response tests | Weeks 4, 6, 8 |
| I. 8 | 100% of approvals, role changes, account restrictions, moderation actions, and sponsorship acceptances create an audit record with actor, action, target, and time. | Automated tests and audit-log review | Weeks 6, 8 |
| I. 9 | The report → hide → clear or remove workflow works end to end, including in a test environment with no UPRM office or department organization. | End-to-end test | Weeks 6, 8 |
| I. 10 | Company-originated route works end to end; the company receives only the data the student chose to share. | End-to-end test and demo | Weeks 6, 8 |
| I. 11 | Student-originated route works end to end; 0 private contact fields are exposed before acceptance. | End-to-end test and demo | Week 8 |
| I. 12 | 100% of required notifications (verification code, Express Interest received, request accepted or declined) are sent in test runs. | Automated tests and email-service logs | Weeks 2, 6, 8 |
| I. 13 | At least 20 core workflow scenarios complete with 0 unhandled server errors, and all required automated tests pass in CI. | Acceptance suite, CI, server logs | Week 8 |
| I. 14 | 0 unresolved High or Critical findings in the automated security scan. | OWASP ZAP scan | Weeks 4, 6, 8 |
| I. 15 | 0 unresolved Critical or Serious accessibility findings (WCAG 2.2 AA), and all primary workflows can be completed using only the keyboard. | axe-core and manual keyboard check | Weeks 4, 6, 8 |
| I. 16 | After a restart, the service is healthy within 5 minutes with no data loss. | Recovery test | Week 8 |
| I. 17 | At least 6 representative users complete role-appropriate tasks with at least 80% overall task success. | Moderated usability sessions | Week 8 |

# 4. Solution Approach

## 4(a) Proposed Solution

**The proposed solution is a web application organized into five connected parts:**

- **Access and trust.** UPRM members verify their institutional email, institutional roles are authorized by administrators, and companies must be approved before they can publish. Every listing shows a source label.
- **Structured publishing.** Each of the six opportunity types has its own form and required fields.
- **Discovery and responses.** Students search, filter, save, and respond to listings. A publisher sees only the students who respond to its own listings.
- **Two-sided capstone sponsorship.** Companies publish project calls; student teams publish projects seeking support and decide whether to accept a company's Express Interest.
- **Moderation and accountability.** Users report suspicious listings; moderators review reports, hide or remove listings, and restrict accounts; every administrative action is recorded in an audit log.

*Example:* a company registers, is approved, publishes an internship, and receives responses from interested students, without ever being able to browse the student population. In the other direction, a student team publishes its capstone project and decides whether to accept a company's interest. The technical design is described in Section 5.

## 4(b) Gap with Existing Solutions

**The gap is that no existing tool combines all six opportunity types for the UPRM community under one trust and privacy model.** Each existing tool covers only one part (Section 5(c) compares them in detail):

- **UPRM offices and departments** publish useful opportunities, but each through its own channel.
- **Handshake** supports employment recruiting, with employer approval managed by each partner school [15], but it does not cover research participation or two-sided capstone sponsorship.
- **Sona Systems** is specialized in research-participant recruitment only [16].
- **LinkedIn and Indeed** are general employment platforms, not built around one campus community [17], [18].

### 4(c) Value Proposition

**The value of Colegio Connect is one trusted path from finding an opportunity to making a controlled connection.** Students search across all opportunity types without needing to know which office owns them. Publishers reach verified UPRM students through one structured process. Student information is shared only through an action the student chooses. The platform serves a specific niche that general platforms do not: a single campus community with employment, research, and capstone opportunities in one place.

## 4(d) Benefits and Limitations

| Audience | Benefit |
|---|---|
| Students | Less time checking separate channels; one place to search and save; clear source information; control over personal data. |
| UPRM Faculty / Offices / Departments | A structured way to publish opportunities with a visible UPRM identity, and a clear list of interested students. |
| External Companies | A controlled way to reach verified UPRM students and to sponsor student capstone projects. |
| Capstone Teams | A way to publish a project, describe the support needed, and choose which company connections proceed. |
| Admins / Moderators | One approval, reporting, and audit process for all opportunity types. |

**Table 5. Benefits by audience.**

**Limitations compared with existing solutions.** Colegio Connect focuses on discovery and controlled connection for UPRM, so existing tools will remain better in their own areas, and the project will not try to compete with them there:

- **Handshake and LinkedIn** will remain better for national employer networks, career fairs, application tracking, and professional networking beyond UPRM.
- **Sona Systems** will remain better for managing research studies, including time-slot scheduling and participation credit.
- **Colegio Connect does not manage applications or interviews.** Students apply through the publisher's own process.
- **Coverage depends on participation**, and moderation reduces but cannot eliminate misleading listings (Section 2(b)).

## 4(e) Resources Required

| Resource | What is needed | Cost / availability |
|---|---|---|
| Team | Three student developers for eight weeks. | Team time |
| Guidance | Advisor feedback at sprint reviews and scope decisions. | No cost |
| Test users | At least six volunteer participants for usability testing. | Volunteers |
| Libraries and frameworks | Java 25, Spring Boot, Spring Security, Spring Data JPA, Hibernate, Flyway, React, TypeScript, springdoc-openapi. | Open source |
| Testing tools | JUnit, Mockito, Testcontainers, Playwright, k6, OWASP ZAP, axe-core. | Open source |
| Development tools | IDEs (IntelliJ IDEA Community Edition, Visual Studio Code), Git, Docker. | Free |
| Development services | GitHub repository, GitHub Actions CI, and Dependabot alerts. | Free / academic tier |
| Hosting | Container host and hosted PostgreSQL able to support the test workload. | Free or low-cost tier |
| Email delivery | Transactional email service for verification codes and notifications. | Free or low-cost tier |
| Accounts and credentials | Hosting and email API keys, and UPRM email test accounts for verification testing. | Free |
| Equipment | Team laptops able to run the Docker development environment. No GPU or special hardware is needed. | Already available |

**Table 5. Benefits by audience.**

The project is designed for a near-zero budget and does not depend on a single hosting or email vendor. Operating the platform after the semester would require an owner, a moderator, recurring hosting and email costs, backups, and security updates. The team and advisor will decide who assumes these responsibilities before handoff.

### 4(f) Intellectual Property

**Intellectual property used.** The project uses open-source software, including Spring Boot (Apache License 2.0) [19], PostgreSQL (PostgreSQL License) [20], and React (MIT License) [21]. The team will keep the required license notices and review all dependency licenses before final delivery. Third-party services, including GitHub, the hosting provider, the email provider, and external form platforms such as Google Forms or Qualtrics, remain the property of their owners and are used under their terms of service.

**Intellectual property generated.** Source code, database schemas, API definitions, diagrams, tests, and documentation are project outputs. If the project generates intellectual property that could be protected, the team will follow the channels provided by the University, together with the course requirements and the advisor's guidance.

**User content and capstone projects.** Listing descriptions, organization information, and student project descriptions remain the property of whoever submits them; the platform only needs permission to store and display them. Publishing a project or accepting Express Interest does not transfer ownership of a student project. Any sponsorship, confidentiality, or licensing terms are handled by a separate agreement, such as those used in Open Capstone [10].

**UPRM names and marks.** The prototype will follow UPRM guidelines for institutional marks [22] and will not present itself as an official UPRM service without authorization.

## 5. Technical Approach

### 5(a) System Architecture Framework

**Colegio Connect uses a client-server architecture with a layered backend** (Figure 1). A React [23] and TypeScript [24] single-page application runs in the browser and communicates over

HTTPS with a REST API that exchanges JSON. The backend is a Java 25 [25] and Spring Boot [26] application organized in layers: REST controllers, a security layer, services, and repositories. Data is stored in PostgreSQL [20]. The backend runs in a Docker container [27], and a transactional email service sends verification codes and notifications. External forms, such as Google Forms or Qualtrics, are reached by link when a research-participation listing uses one.

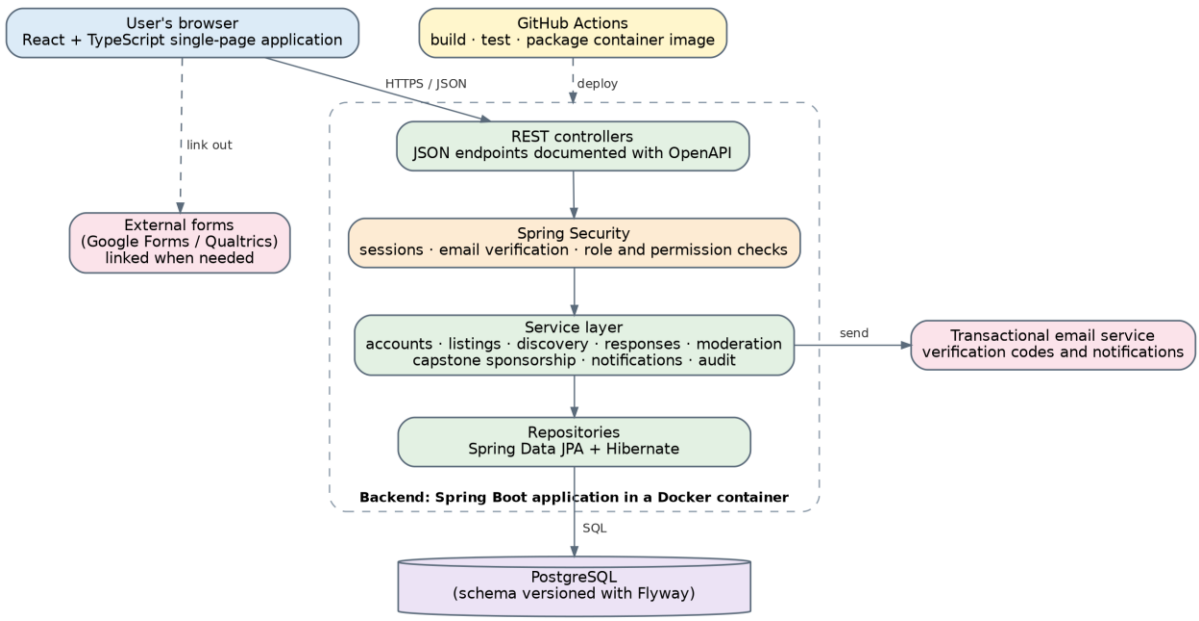



*Figure 1. System architecture.*

**Modules.**

Table 7 describes the main modules of the system.

| Module | Responsibility |
|---|---|
| Web client | React and TypeScript interface for all four roles; calls the REST API and shows listings, forms, and dashboards. |
| Accounts and verification | Registration, institutional email verification by code, login, sessions, and password handling. |
| Organizations and roles | Organization records, company approval, role assignments, and organization membership. |
| Listings | Creation, editing, source labels, and lifecycle of the six opportunity types (Figure 4). |
| Discovery | Feed, keyword search, filters, and saved listings. |
| Responses | Student responses to listings and the publisher's view of its own respondents. |
| Capstone sponsorship | Student teams, team projects, Express Interest requests, and consent-based contact release. |

| Module | Responsibility |
|---|---|
| Moderation | Reports, review queues, hiding and removal of listings, and account restrictions. |
| Notifications | Sends required emails through the email service. |
| Audit | Records who did what, to which record, and when, for every administrative and sponsorship action. |

**Table 7. System modules.**

| Layer | Technology | Role in the system |
|---|---|---|
| Frontend | React + TypeScript | Single-page interface for all roles. |
| Backend | Java 25 LTS + Spring Boot 4.1 | REST API, business rules, moderation, sponsorship, and notifications. |
| API | REST over HTTPS, JSON payloads | Request-response contract between frontend and backend. |
| Security | Spring Security | Authentication, server-side sessions, CSRF protection, and role checks; ready for OAuth 2.0 / OpenID Connect. |
| Database | PostgreSQL 18 | Relational storage for all system records. |
| Persistence | Spring Data JPA + Hibernate | Maps entities to tables using parameterized queries. |
| Migrations | Flyway | Versioned schema changes across all environments. |
| API documentation | springdoc-openapi | Generates the OpenAPI contract and interactive documentation. |
| Containers | Docker | Same runtime for development, CI, and deployment. |
| Testing | JUnit, Mockito, Testcontainers, Playwright, k6, OWASP ZAP, axe-core | Unit, integration, end-to-end, load, security, and accessibility testing. |
| CI/CD | GitHub Actions | Builds and tests every push and pull request. |

**Table 8. Technology stack.**

**Why this architecture.** Layering keeps high-risk logic, such as authorization, moderation, and auditing, in the backend instead of the interface. A relational database fits the data, because users, roles, organizations, listings, and responses have clear relationships. TypeScript and Spring Boot are established tools with strong support for security, validation, and testing, which suits a three-person team. Docker keeps the system portable between the team's laptops, CI, and the host. The prototype verifies users by institutional email, but the security layer can later connect to a UPRM identity provider through OAuth 2.0 / OpenID Connect without a redesign.

**Access model.** There are two account types. UPRM_MEMBER accounts verify an institutional email and can be assigned the Student, Faculty/Office/Department, or Admin/Moderator roles. EXTERNAL_ORGANIZATION accounts register with organization details and a business email, and receive company capabilities only after administrator approval. Every person has an individual login; offices, departments, and companies are separate organization records linked to their authorized members. A user may hold more than one role, for example a professor who is also a moderator.

**Source labels.** Every listing shows a source label, which also decides how it is reviewed (Table 9 and Figure 4).

| Label | Who posts | Review |
|---|---|---|
| Verified UPRM institution | Authorized faculty, office, or department. | Published directly unless the poster is flagged; remains reportable. |
| Approved external company | Approved company. | The first listings are reviewed before publication; after a set number of listings with no confirmed violations, later listings publish directly and remain reportable. |
| Verified student project | Verified student team. | Reviewed before publication; private contact data is never part of the public listing. |

**Table 9. Source labels and review rules.**

## 5(b) Diagrams

Four diagrams describe the key aspects of the system: the architecture (Figure 1), the data model (Figure 2), the use cases by role (Figure 3), and the listing lifecycle (Figure 4). Figures 3 and 4 use UML 2.5.1 notation [28].

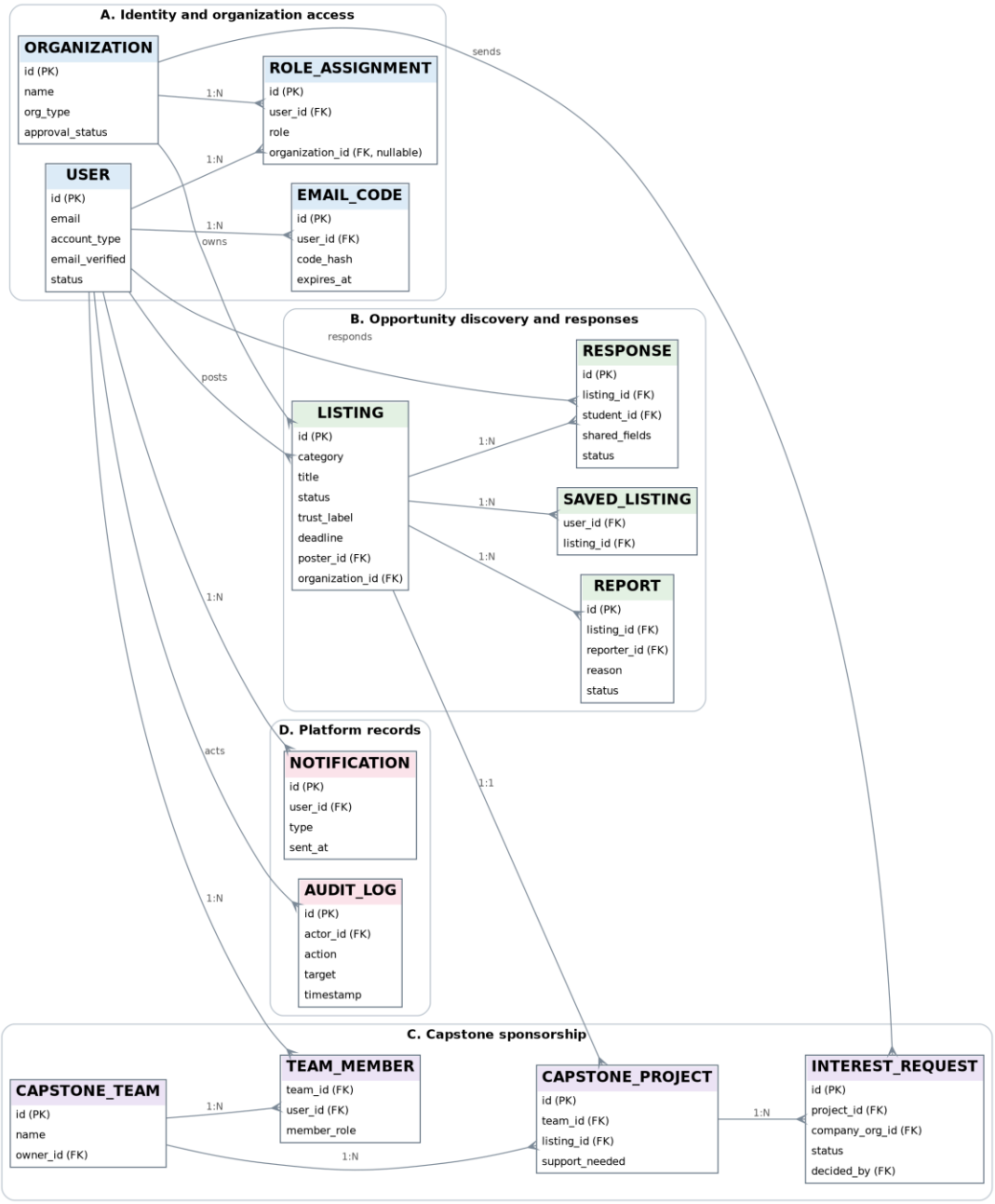



*Figure 2. Simplified entity-relationship model, grouped by function (PK = primary key, FK = foreign key; the forked end of each line marks the "many" side).*

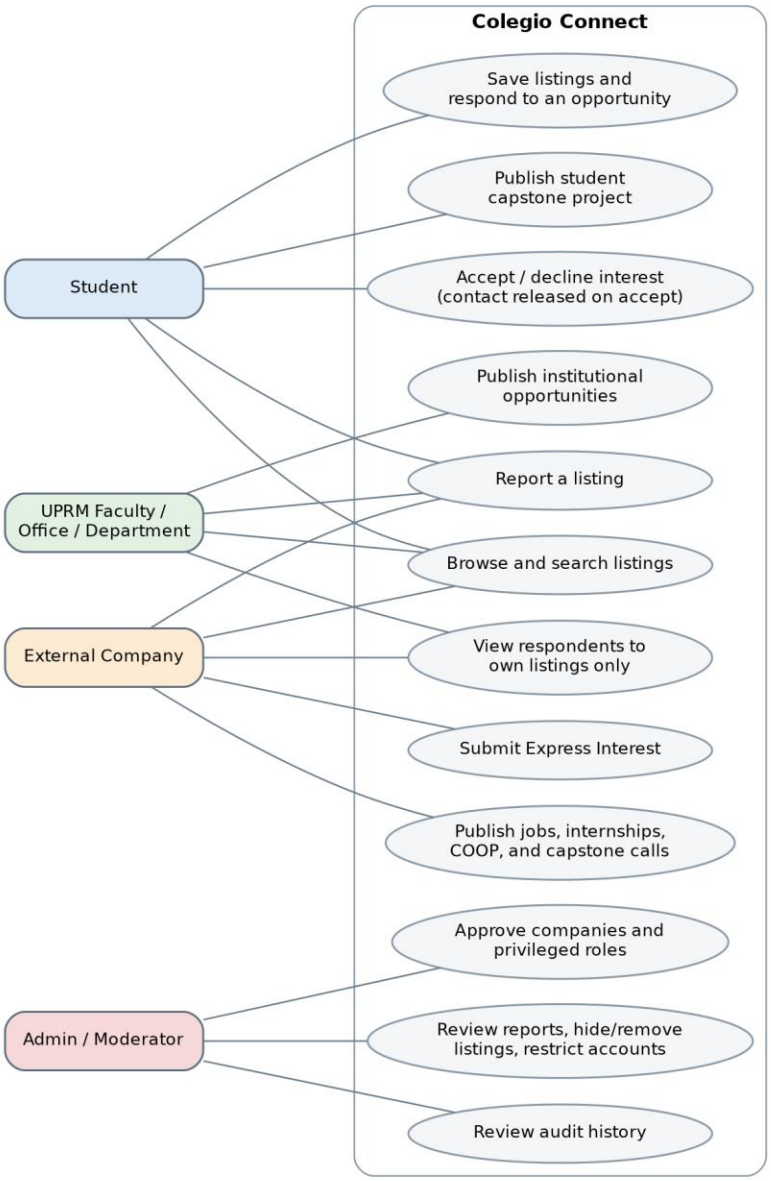



*Figure 3. Use cases by role. Admin/Moderator is an additional role that may be held by a user who also has another role.*

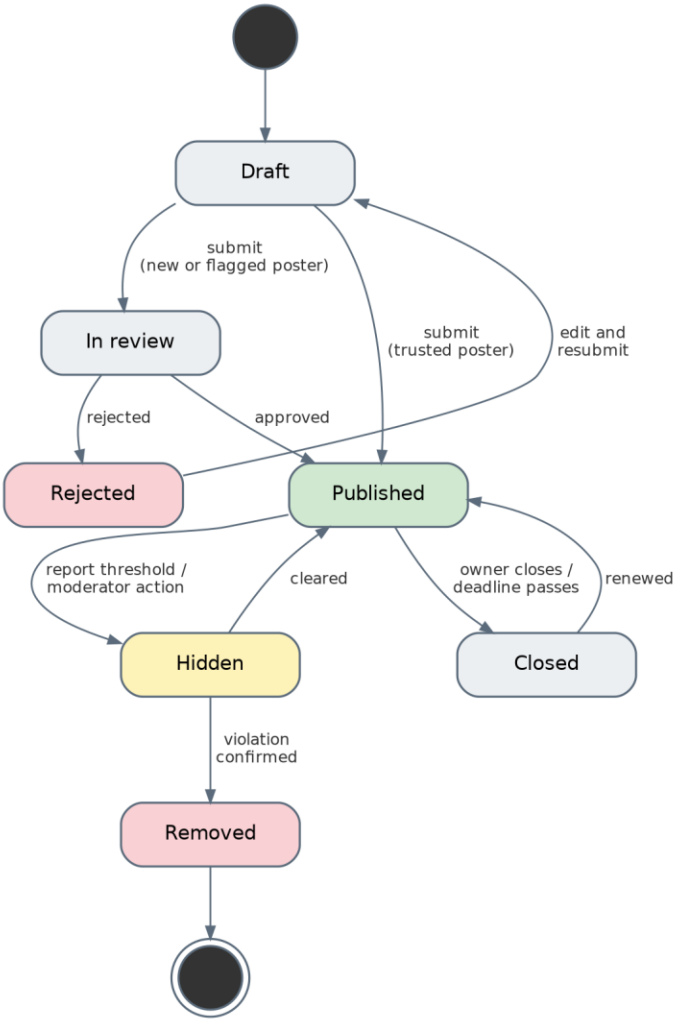



*Figure 4. Listing lifecycle state diagram.*

**Reading Figure 4.** A listing from a new or flagged poster goes to review before publication, while a trusted poster may publish directly. A published listing can be closed by its owner or when its deadline passes, and can be renewed. If reports or a moderator require it, the listing is hidden until an administrator clears it or confirms a violation and removes it. Every administrative state change is written to the audit log.

## 5(c) Related Works

**Existing systems.** Table 10 compares the systems that solve parts of the same problem.

| System | Jobs / intern. / COOP | Research positions | Research participation | Capstone sponsorship | Trust / moderation |
|---|---|---|---|---|---|
| UPRM Placement Office [6] | Yes | No | No | No | Managed by the office |
| ADEM + Handshake [7], [15] | Yes | Varies | No | No | Employer approval by each school; used at UPRM [29] |
| Sona Systems [16], [30] | No | No | Yes | No | Participant-pool administration |
| LinkedIn [17] | Yes | No | No | No | General platform rules |
| Indeed [18] | Yes | No | No | No | General employer controls |
| Capstone Marketplace [14] | No | No | No | Yes, one-sided | Course-managed |
| **Colegio Connect (proposed)** | **Yes** | **Yes** | **Yes (discovery only)** | **Yes, two-sided** | **Roles, source labels, reports, moderation, audit** |

**Table 10. Comparison with existing systems.**

Handshake is the closest comparison for employment, Sona Systems for research participation, and LinkedIn and Indeed are general job platforms. The previous UPRM projects PRJobs [12] and the UPRM Research Portal [13] each addressed one opportunity type. None of them combines all six opportunity types with one trust, privacy, and moderation model and two-sided capstone sponsorship.

**Existing tools reused.** Instead of building common functions from scratch, the project reuses established tools (Table 11).

| Tool | What it provides | What the team does not need to build |
|---|---|---|
| Spring Security [31] | Authentication, sessions, password hashing, CSRF protection, authorization rules. | Custom login, session, and password code. |
| Spring Data JPA [32] + Hibernate [33] | Object-relational mapping and parameterized queries. | Hand-written SQL access layer. |
| Flyway [34] | Versioned database migrations. | Manual schema updates. |
| springdoc-openapi [35] | OpenAPI contract and interactive API documentation generated from the code. | Manually written API documentation. |
| Testcontainers [36] | Disposable PostgreSQL containers for integration tests. | Shared test databases. |
| JUnit [37] + Mockito [38] | Unit testing and mocking. | Custom test harness. |
| Playwright [39] | Automated browser tests for end-to-end workflows. | Manual repetition of workflows. |
| k6 [40] | Scripted load tests with concurrent virtual users. | Custom load generator. |
| OWASP ZAP [41] | Automated web security scanning. | Custom security scanner. |
| axe-core [42] | Automated accessibility checks. | Manual-only accessibility review. |
| GitHub Actions [43] + Dependabot [44] | CI pipelines and dependency vulnerability alerts. | Custom build server and manual dependency checks. |

**Table 11. Existing tools reused by the project.**

# 5(d) Realistic Engineering Constraints

**Table 12. Engineering constraints.**

| # | Constraint | Justification | Design implication |
|---|---|---|---|
| C1 | Eight-week schedule | Prototype, testing, documentation, and presentation must fit a short period. | Four two-week sprints with a defined order for reducing scope (Section 6(a)). |
| C2 | Three developers | Design, coding, testing, and documentation are shared by three people. | Work split by feature; an OpenAPI contract allows parallel frontend and backend work; every change is reviewed. |
| C3 | Near-zero budget | No funding for licenses or infrastructure. | Open-source software and free or low-cost hosting, database, CI, and email services. |
| C4 | Response time on low-cost hosting | Students expect fast search, but the prototype runs on limited resources. | Database indexes on search and filter fields, paginated results, and a 2-second target for 95% of discovery requests (I3). |

| # | Constraint | Justification | Design implication |
|---|---|---|---|
| C5 | No UPRM IT integration | Single sign-on or directory access cannot be assumed for a student project. | Verification by institutional email; security layer ready for OAuth 2.0 / OpenID Connect. |
| C6 | Student data privacy | The system stores student responses and contact information. | Data minimization, HTTPS only, server-side authorization, and consent-based disclosure. |
| C7 | Scam and fraud risk | Fake organizations or misleading listings could be posted. | Company approval, source labels, reports, review queues, removal, restrictions, and audit records. |
| C8 | Unofficial status | The prototype is not automatically an official UPRM service. | Works without any office partnership and clearly shows each listing's source. |
| C9 | Limited moderation capacity | A small team cannot review every listing manually. | Pre-publication review only for new or flagged posters; everything else remains reportable. |
| C10 | Dependence on external email delivery | Verification and notifications rely on a third-party email service with rate limits. | Short-lived hashed codes, resend limits, and logging of delivery results. |

# 5(e) Engineering Standards

**Table 13. Engineering standards.**

| Standard | Applies to | How it is used and why |
|---|---|---|
| OWASP ASVS 5.0, Level 1 [45] | Authentication, access control, input validation, sessions | Security requirements checklist for the modules that protect accounts, roles, and student data. |
| OWASP Top 10 [46] | Whole web application | List of the most critical web risks used to guide code review and the ZAP scan. |
| NIST SP 800-63B-4 [47] | Passwords and verification codes | Guides password length rules, hashed storage, short-lived codes, and rate limiting. |
| WCAG 2.2, Level AA [48] | Web interface | Accessibility baseline for keyboard access, labels, focus, and contrast (I15). |
| RFC 9110, HTTP Semantics [49] | REST API | Correct use of HTTP methods and status codes in every endpoint. |
| RFC 8259, JSON [50] | API payloads | Data format for all requests and responses. |
| OpenAPI Specification 3.1 [51] | REST API | Machine-readable contract between frontend and backend, used for documentation and contract tests. |
| RFC 8446, TLS 1.3 [52] | All client-server traffic | Encrypts credentials, student data, and administrative actions in transit. |
| RFC 6749, OAuth 2.0 [53] and OpenID Connect Core 1.0 [54] | Future identity-provider integration | Keeps the design ready for UPRM single sign-on without making it a prototype dependency. |

| Standard | Applies to | How it is used and why |
|---|---|---|
| ISO/IEC 25010:2023 [55] | Quality requirements | Quality model used to define the security, performance, usability, and reliability targets. |
| ISO/IEC/IEEE 29119-1:2022 [56] | Testing | Common vocabulary and concepts for test levels, test cases, and the test plan in Section 6(d). |
| UML 2.5.1 [28] | Design documentation | Standard notation for the use-case and state diagrams. |
| The Scrum Guide (2020) [57] | Project management | Defines the sprint events, roles, and artifacts the team follows. |

Whether FERPA and specific UPR data-handling policies apply to an independent student prototype will be confirmed with the advisor and, when possible, UPRM personnel. The proposal does not claim compliance that has not been formally established.

## 5(f) Best Practices

**Table 14. Best practices.**

| Practice | How it is applied | Risk reduced |
|---|---|---|
| Scrum with two-week sprints [57] | Sprint planning, a weekly progress check, sprint review, and a short retrospective every two weeks. | Scope creep and late discovery of problems. |
| Pull requests and code review | All work is done in branches; merging to main requires approval from another member. | Defects and unreviewed changes. |
| Continuous integration [59] | GitHub Actions runs the test suite on every push and pull request; failing checks block merging. | Regressions and broken builds. |
| Test pyramid [60] | Many unit tests, fewer integration tests, and a small set of end-to-end tests. | Slow or unreliable test suites. |
| Server-side least privilege | Every protected endpoint checks role, account status, membership, ownership, and consent. | Unauthorized access to student or admin data. |
| Secure credential handling | Hashed passwords, secrets in environment variables, and rate-limited login and verification. | Credential theft, brute force, and leaked keys. |
| Database migrations | Flyway versions every schema change. | Inconsistent databases across environments. |
| Synthetic test data | Development and testing use only generated data. | Exposure of real student information. |
| Audit logging | Administrative and sponsorship actions record actor, action, target, and time. | Unaccountable decisions. |
| Pair programming for high-risk logic [61] | Authorization and contact-release code is written or reviewed in pairs. | Security defects in critical code. |
| Dependency management | Dependencies are pinned and monitored with Dependabot alerts. | Known vulnerable packages. |

# 6. Project Plan

## 6(a) Activities and Process

**The team follows Scrum with four two-week sprints** [57]. Each sprint includes planning, implementation, testing, a weekly progress check, a review of the working result with the advisor when possible, and a short retrospective. At the end of each sprint the team compares completed work against the indicators in Section 3(b), updates the backlog, and records any scope decision.

**Definition of done.** A task is done only when its code has been reviewed, its tests pass in CI, and the OpenAPI document reflects any API change.

**Scope rule.** If the schedule slips, the team will reduce, in this order: extra notifications beyond the required ones, optional listing fields, nonessential interface improvements, and analytics. Access control, discovery, saving, responses, privacy, moderation, required notifications, and both capstone routes are not reduced. Any larger change will be reviewed with the advisor.

**Initial product backlog.** Table 15 lists the main user stories known today. The backlog will be refined during the project.

**Table 15. Initial product backlog.**

| ID | User story | Sprint |
|---|---|---|
| US1 | As a UPRM member, I want to register with my institutional email and verify it with a code, so that I receive UPRM permissions. | 1 |
| US2 | As an administrator, I want to assign institutional and moderator roles, so that only authorized people publish or moderate. | 1 |
| US3 | As a company, I want to register my organization and request approval, so that I can publish opportunities. | 2 |
| US4 | As an administrator, I want to approve or reject company requests, so that only legitimate organizations publish. | 2 |
| US5 | As a publisher, I want to create a listing in any of the six categories, so that students can find it. | 2 |
| US6 | As a student, I want to search and filter listings by category, keyword, source, organization, and deadline, so that I find relevant opportunities quickly. | 2 |
| US7 | As a student, I want to save listings, so that I can return to them later. | 2 |
| US8 | As a student, I want to respond to a listing and choose which profile fields to share, so that I control my information. | 3 |

| ID | User story | Sprint |
|---|---|---|
| US9 | As a publisher, I want to see the students who responded to my listing, so that I can contact them. | 3 |
| US10 | As a user, I want to report a suspicious listing, so that moderators can review it. | 3 |
| US11 | As a moderator, I want to hide or remove listings and restrict accounts, with each action recorded, so that the platform stays safe and accountable. | 3 |
| US12 | As a company, I want to publish a capstone project call, so that student teams can respond. | 3 |
| US13 | As a student team, I want to publish our project and the support we need, so that sponsors can find us. | 4 |
| US14 | As a company, I want to send Express Interest to a student project, so that I can offer support. | 4 |
| US15 | As a team member, I want to accept or decline a company's interest, so that our contact information is shared only if we agree. | 4 |

## 6(b) Timetable, Milestones, and Deliverables

Week 1 begins when the proposal is approved, and all work finishes before the end of the Fall 2026 semester.

**Table 16. Timetable, milestones, and deliverables.**

| Sprint | Weeks | Main activities | Milestone and deliverables |
|---|---|---|---|
| 1 | 1–2 | Repository, CI, Docker, hosted PostgreSQL, Flyway baseline, OpenAPI skeleton, wireframes; email verification, login, roles, organizations (US1–US2). | **M1 – Access foundation.** Running frontend, API, and database; working login and role checks; first authorization tests. |
| 2 | 3–4 | Company registration and approval; listing creation, source labels, and lifecycle; feed, search, filters, saving (US3–US7); set up the synthetic workload, load tests, ZAP scan, and axe checks. | **M2 – Publishing and discovery.** Authorized users publish listings; students browse, search, filter, and save; first performance, security, and accessibility results. |
| 3 | 5–6 | Student responses and respondent view; privacy rules; reports, moderation, restrictions, audit log; required notifications; company capstone calls (US8–US12); start the student-team model; repeat scans, tune queries, rehearse recovery, prepare usability sessions. | **M3 – Responses and moderation.** Response flow, moderation and audit flow, notifications, and company capstone route; updated test results and usability plan. |
| 4 | 7–8 | Student capstone projects, Express Interest, accept or decline, contact release (US13–US15); final acceptance tests; fix findings; final documentation. | **M4 – Sponsorship and final acceptance.** Two-sided sponsorship workflow; final results for I1–I17; user guide, final report, and presentation. |

## 6(c) Team Responsibilities

Each member has a primary area and leads selected features from start to finish, while the other two support or review them, so no module depends on only one person:

- **Cristian J. Massini Pérez – Backend and data:** database schema, API, accounts, listings, responses, and capstone sponsorship.
- **Gabriel E. E. Rodríguez Rodríguez – Infrastructure, security, and moderation:** CI/CD, Docker, hosting, authorization, moderation, audit, and security and performance testing.
- **Luis A. Marrero Morales – Frontend and user experience:** interface, discovery screens, notifications, end-to-end tests, accessibility, and usability testing.

**Table 17. Feature leads by sprint.**

| Sprint | Feature | Lead | Support |
|---|---|---|---|
| 1 | Database schema, Flyway, registration, verification, login, roles | Cristian | Gabriel (security review), Luis (screens) |
| 1 | Environment, CI pipeline, Docker, hosted PostgreSQL | Gabriel | Cristian (database), Luis (CI checks) |
| 1 | Wireframes, frontend setup, login and verification screens | Luis | Cristian (API contract), Gabriel (review) |
| 2 | Company approval, institutional roles, source labels, listing lifecycle; ZAP setup | Gabriel | Cristian (role checks), Luis (screens) |
| 2 | Listing creation and publisher screens | Cristian | Luis (components), Gabriel (API review) |
| 2 | Feed, search, filters, saved listings; k6 and axe setup | Luis | Gabriel (queries), Cristian (API review) |
| 3 | Student responses and privacy rules | Cristian | Gabriel (authorization review), Luis (screens) |
| 3 | Reports, moderation, restrictions, audit log; recovery rehearsal | Gabriel | Luis (end-to-end tests), Cristian (role checks) |
| 3 | Notifications and company capstone route; usability preparation | Luis | Cristian (API), Gabriel (privacy tests) |
| 4 | Student projects, Express Interest, accept or decline, contact release | Cristian | Luis (screens), Gabriel (security review) |
| 4 | Final security, performance, and recovery tests; hardening | Gabriel | Cristian (fixes), Luis (verification) |
| 4 | Final usability and accessibility tests; report, presentation, demo | Luis | Cristian and Gabriel |

**Shared responsibilities.** Every pull request is reviewed by at least one other member. The Scrum Master and QA lead roles rotate each sprint. Each feature has a backup member who understands its API, data, and tests.

## 6(d) Testing Plan

**Testing runs at the same time as development.** Tests are written with each feature and follow the test pyramid [60]: many JUnit and Mockito unit tests, fewer Test containers integration tests against a real PostgreSQL database, and a small set of Playwright end-to-end tests for the critical workflows. Test terms follow ISO/IEC/IEEE 29119-1 [56].

- **On every push and pull request:** GitHub Actions runs all unit and integration tests; failing tests block merging.
- **At the end of every sprint:** the full suite, including end-to-end tests, runs against the deployed build, and the results for the indicators covered so far are recorded.
- **From Sprint 2 on:** the k6 load test, OWASP ZAP scan, and axe-core checks run at the end of each sprint on the features implemented so far.
- **In Week 8:** the final acceptance suite verifies every indicator (I1–I17) against its full target.

| Requirement | Test type and tool | Acceptance criterion | Sprint |
|---|---|---|---|
| UPRM email verification | Integration (JUnit, Testcontainers) | I5: invalid emails and codes rejected; rate limit enforced. | 1–4 |
| Role-based access | Integration (Testcontainers) | I6: 100% of at least 10 negative cases pass. | 1–4 |
| Student data privacy | API-response tests | I7: 0 private fields in unauthorized responses. | 2–4 |
| Listing categories and lifecycle | Integration + end-to-end (Playwright) | I1: all 6 categories; state changes follow Figure 4. | 2–4 |
| Search and filters | Functional tests | I2: 18 cases, 100% correct. | 2–4 |
| Discovery performance | Load test (k6) | I3: 95% under 2 s, errors below 1%. | 2–4 |
| Saved listings | Integration | I4: save, unsave, revisit in all 6 categories. | 2–4 |
| Audit records | Integration + log review | I8: 100% of listed actions audited. | 3–4 |
| Moderation workflow | End-to-end (Playwright) | I9: report → hide → clear or remove, with no office organization present. | 3–4 |
| Company capstone route | End-to-end (Playwright) | I10: company sees only shared data. | 3–4 |
| Student capstone route | End-to-end (Playwright) | I11: 0 contact fields before acceptance. | 4 |
| Notifications | Integration + email logs | I12: 100% of required notifications sent. | 1–4 |

| Requirement | Test type and tool | Acceptance criterion | Sprint |
|---|---|---|---|
| Overall stability | Acceptance suite + CI | I13: 20 scenarios, 0 unhandled errors. | 4 |
| Security | OWASP ZAP + ASVS checklist | I14: 0 unresolved High or Critical findings. | 2–4 |
| Accessibility | axe-core + manual keyboard check | I15: 0 Critical or Serious findings. | 2–4 |
| Recovery | Restart test | I16: healthy within 5 min, no data loss. | 3–4 |
| Usability | Moderated sessions | I17: 6 users, at least 80% task success. | 3 (prep), 4 |

**Table 18. Tests for functional and quality requirements.**

**Compliance with engineering constraints.** Table 19 shows how each constraint from Section 5(d) is checked.

| Constraint | How compliance is verified |
|---|---|
| C1 Eight-week schedule | Sprint reviews compare completed work with milestones M1–M4; scope decisions are recorded. |
| C2 Three developers | Every merged pull request has a reviewer; OpenAPI contract tests run in CI. |
| C3 Near-zero budget | Hosting and email costs are recorded at each sprint review. |
| C4 Response time | k6 load test at the end of Sprints 2–4 (I3). |
| C5 No UPRM IT integration | Verification tests run with no external identity provider (I5). |
| C6 Student data privacy | Authorization and API-response tests (I6, I7, I11). |
| C7 Scam and fraud risk | Approval, report, and moderation tests (I6, I9). |
| C8 Unofficial status | Acceptance test with no UPRM office organization (I9). |
| C9 Limited moderation capacity | Lifecycle tests confirm only new or flagged posters go to review (I1). |
| C10 Email delivery | Notification tests and delivery logs (I12). |

**Table 19. Verification of engineering constraints.**

# 7. Previous Work and References

## 7(a) Previous Work

Two previous UPRM CSE Capstone projects are directly related. PRJobs focused on job search and company job postings [12], and the UPRM Research Portal connected students and professors around research opportunities [13]. Outside UPRM, the Capstone Marketplace showed that an online tool can match capstone teams with industry sponsors [14]. Colegio Connect builds on the need shown by these projects and combines employment, research, research participation, and two-sided capstone sponsorship under one shared access, privacy, and moderation model.

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